

četrverec hitrosti:

$\frac{d\mathbf{r}}{dt}$ ne bo v redu, ker t koordinatni čas

lastni čas τ :

$$\Delta\tau = \int_{t_1}^{t_2} \frac{dt}{\gamma}$$

$$\underline{v} = \frac{d\mathbf{r}}{d\tau} = \left(c \frac{dt}{d\tau}, \frac{dx}{d\tau}, \frac{dy}{d\tau}, \frac{dz}{d\tau} \right) =$$
$$= \left(c\gamma, \gamma\vec{v} \right) \rightarrow \underline{v}' = \underline{L} \underline{v}$$

$$\rightarrow \underline{p} = m \underline{v} \rightarrow \underline{\text{četrverec GK}}$$
$$= (m c \gamma, m \gamma \vec{v})$$
$$= \left(\frac{E}{c}, \vec{p} \right) \rightarrow \underline{p}' = \underline{L} \underline{p}$$

zveza med E in GK :

$$\underline{p} = \left(\frac{E}{c}, \vec{p} \right), \text{ je četrverec,}$$

ima invarianto $\underline{p} \cdot \underline{p} (= \underline{p}' \cdot \underline{p}')$

$$\underline{p}: \text{v abs. sist.} \quad \underline{p} = \left(\frac{E}{c}, \vec{p} \right)$$

$$\underline{p}': \text{v sist. delca (lastni sist.)} \quad \underline{p}' = \left(\frac{mc^2}{c}, \vec{0} \right) \quad \swarrow \text{v last. sist. delec nikamor ne gre!}$$

$$\underline{p}' \cdot \underline{p}' = \underline{p} \cdot \underline{p}$$

$$\left(\frac{E_0}{c} \right)^2 - \vec{0}^2 = \left(\frac{E}{c} \right)^2 - \vec{p}^2$$

$$\boxed{E^2 = m^2 c^4 + \vec{p}^2 c^2}$$

→ za brezmasne delce:

$$\boxed{E = pc}$$

↙ foton (kvant EM valovanja)
↘ neutrino (skoraj brezmasni)
se giblje s c

